

AI in Real World Context

A 60-minute field guide to tools, services and next-token predictors

Shuvam Banerji
Seal

BS–MS final year, IISER Kolkata

github.com/Shuvam-Banerji-Seal

August 2026

▷ 1. Prologue: AI in the Real World

- Real → Current
- The Question

▷ 2. The Agenda

- Reading the Room
- The Goal

▷ 3. Tool Time

- What Kind of Tools?
- Are All These AI Tools?

▷ 4. The AI Landscape

- The Umbrella Term
- Next-Token Predictors
- The Lens & The Assistant Axis
- The Alignment Wall

▷ 5. Fundamentals

- Services
- Side Note
- Where Information Lives
- It's All Text

▷ 6. Knowledge Compression

- Human Knowledge → Model
- The Distribution
- Too Much Maths?

▷ 7. Tools You Should Learn

- Git & Friends
- L^AT_EX & Overleaf
- Do It Yourself?

▷ 8. Agentic Tools

- Use Agentic Tools
- The Definition

1. Prologue: AI in the Real World

AI in Real → Current World Context

- ◇ **Real** promises a fixed world — it isn't one.

AI in Real → Current World Context

- ◇ **Real** promises a fixed world — it isn't one.
- ◇ **Current**: what exists *today*, in your browser, your phone, your lab.

AI in Real → Current World Context

- ◇ **Real** promises a fixed world — it isn't one.
- ◇ **Current**: what exists *today*, in your browser, your phone, your lab.
- ◇ Everything in this talk is something you can touch **this week**.

so, let's start with you

What do you mean by “AI”?

◇ ChatGPT / Gemini / Copilot?

so, let's start with you

What do you mean by “AI”?

- ◇ ChatGPT / Gemini / Copilot?
- ◇ Robots & self-driving cars?

so, let's start with you

What do you mean by “AI”?

- ◇ ChatGPT / Gemini / Copilot?
- ◇ Robots & self-driving cars?
- ◇ The algorithm that recommends your reels?

so, let's start with you

What do you mean by “AI”?

- ◇ ChatGPT / Gemini / Copilot?
- ◇ Robots & self-driving cars?
- ◇ The algorithm that recommends your reels?
- ◇ A calculator on steroids?

so, let's start with you

What do you mean by “AI”?

- ◇ ChatGPT / Gemini / Copilot?
- ◇ Robots & self-driving cars?
- ◇ The algorithm that recommends your reels?
- ◇ A calculator on steroids?
- ◇ Pure hype?

so, let's start with you

What do you mean by “AI”?

- ◇ ChatGPT / Gemini / Copilot?
- ◇ Robots & self-driving cars?
- ◇ The algorithm that recommends your reels?
- ◇ A calculator on steroids?
- ◇ Pure hype?

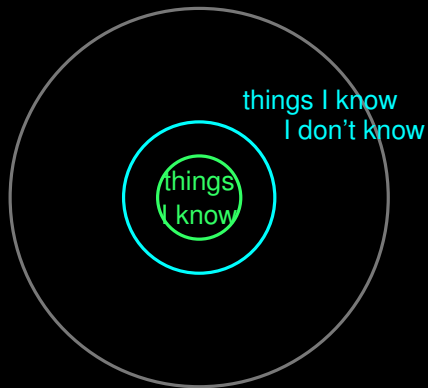
the deal

Whatever you said — we'll build on it; by the end you'll have a sharper answer.

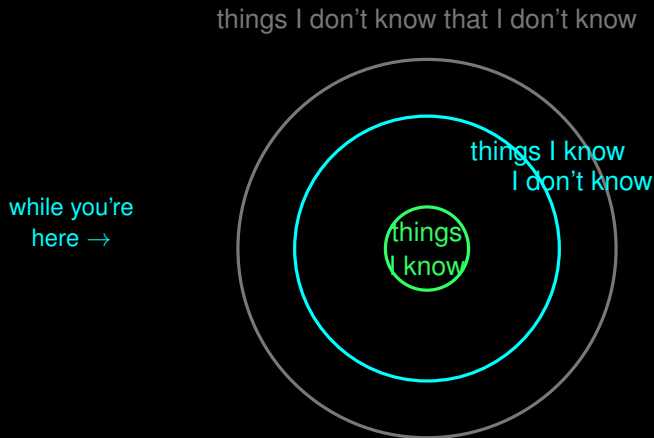
2. The Agenda

The room, mapped | three circles of knowing

things I don't know that I don't know



The room, mapped | three circles of knowing



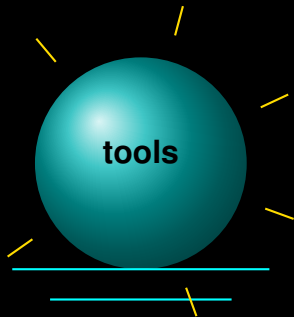
Objective: increase the area of **things you know you don't know**.

The room, mapped | three circles of knowing



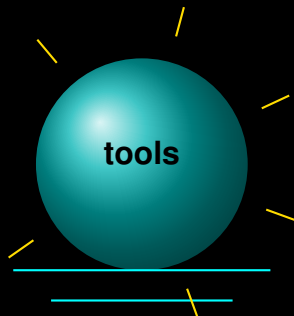
Objective: increase the area of **things you know you don't know**.

I had a premonition | and it was about tools



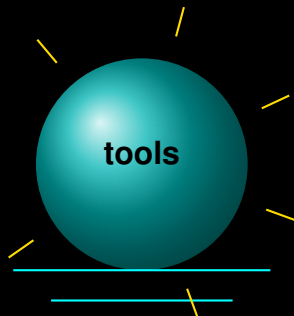
- ◇ The audience wants **tools**, not theory.

I had a premonition | and it was about tools



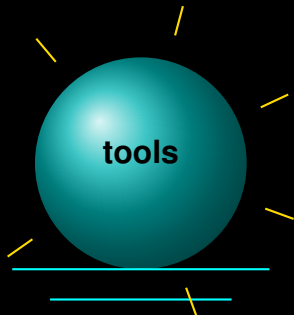
- ◇ The audience wants **tools**, not theory.
- ◇ I got the premonition.

I had a premonition | and it was about tools



- ◇ The audience wants **tools**, not theory.
- ◇ I got the premonition.
- ◇ And the agenda wrote itself:

I had a premonition | and it was about tools



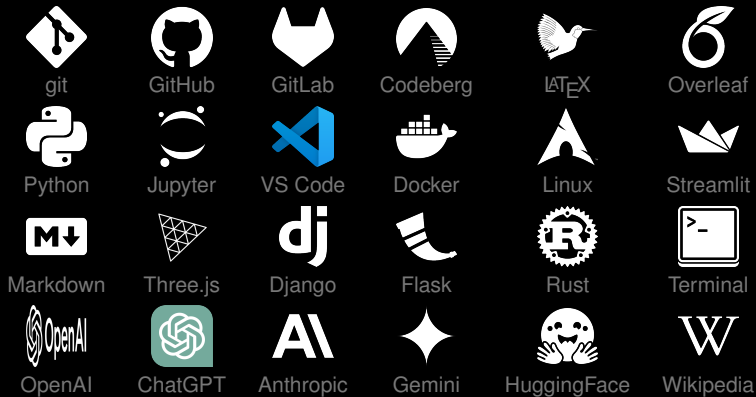
- ◇ The audience wants **tools**, not theory.
- ◇ I got the premonition.
- ◇ And the agenda wrote itself:

Goal #1

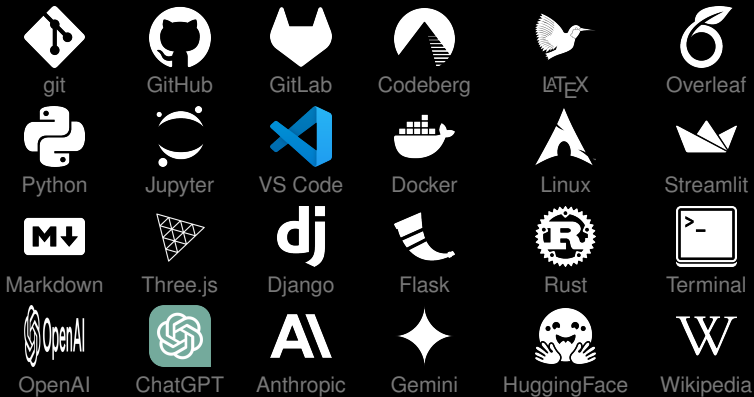
In the next **1 hour**, you will learn about tools.

3. Tool Time

What kind of tools? | spoiler: not all of these are AI



What kind of tools? | spoiler: not all of these are AI



Are all these AI tools?

Most of them aren't AI.

Git, $\text{\LaTeX}$, the terminal — older than most of you, zero AI.
The AI ones (ChatGPT, Gemini, Claude) *sit on top* of these.

Most of them aren't AI.

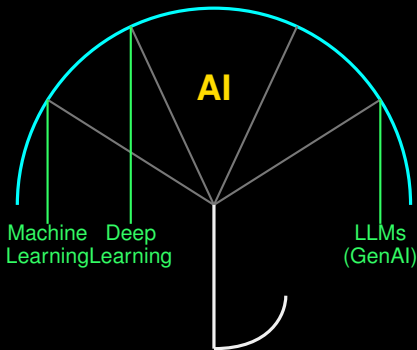
Git, $\text{\LaTeX}$, the terminal — older than most of you, zero AI.
The AI ones (ChatGPT, Gemini, Claude) *sit on top* of these.

the point

The non-AI tools are **leverage**: learn them once, use them forever.
With them, the AI tools become **superpowers** instead of toys.

4. The AI Landscape

AI is an umbrella term | what I work in: LLMs

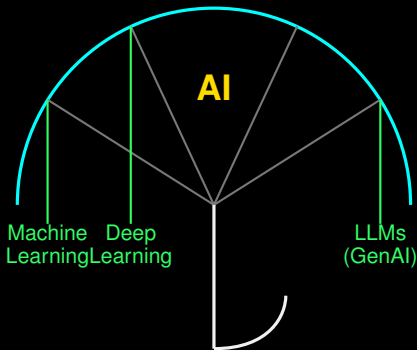


- ◇ **AI** = the umbrella over everything "intelligent".



xkcd 1838 — machine learning, the honest version

AI is an umbrella term | what I work in: LLMs



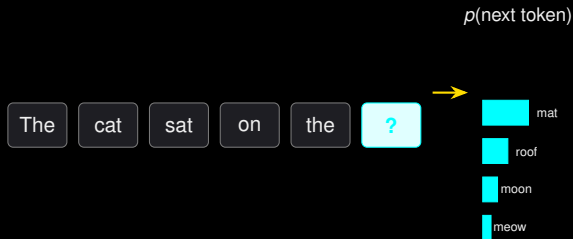
- ◇ **AI** = the umbrella over everything "intelligent".
- ◇ What I work in: **LLMs** — *next-token predictors*.



xkcd 1838 — machine learning, the honest version

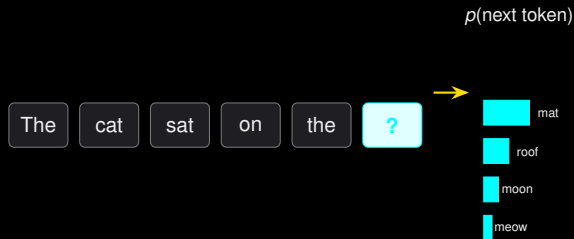
LLMs = next-token predictors | everything ChatGPT does, one loop

- ◇ The model never *searches* — it *predicts*.



Input: a token sequence $\longrightarrow$ **Output:**
probabilities over the next token.

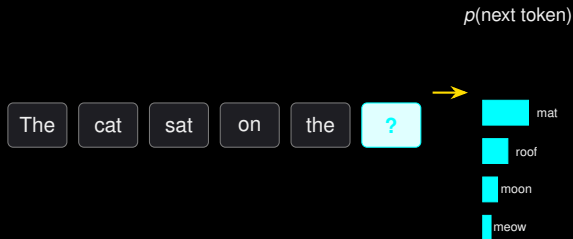
LLMs = next-token predictors | everything ChatGPT does, one loop



- ◇ The model never *searches* — it *predicts*.
- ◇ It repeats this loop, token by token, until you stop it.

Input: a token sequence → **Output:** probabilities over the next token.

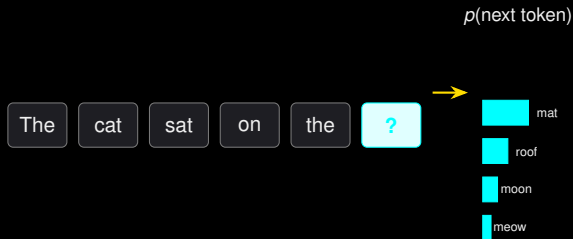
LLMs = next-token predictors | everything ChatGPT does, one loop



Input: a token sequence $\longrightarrow$ **Output:** probabilities over the next token.

- ◇ The model never *searches* — it *predicts*.
- ◇ It repeats this loop, token by token, until you stop it.
- ◇ Add tools, and prediction becomes **action**.

LLMs = next-token predictors | everything ChatGPT does, one loop



Input: a token sequence $\longrightarrow$ **Output:** probabilities over the next token.

- ◇ The model never *searches* — it *predicts*.
- ◇ It repeats this loop, token by token, until you stop it.
- ◇ Add tools, and prediction becomes **action**.



trust me bro — I know this token

Through the lens | once you see it, you can't unsee it

deep learning AGI reasoning
chatbots agents type

**next token
prediction**

$$p(w_t | w_{<t})$$

- ◇ The **lens**: look at any AI through it.

Through the lens | once you see it, you can't unsee it

deep learning AGI reasoning
chatbots agents type

**next token
prediction**

$$p(w_t | w_{<t})$$

- ◇ The **lens**: look at any AI through it.
- ◇ **Agents**? next-token prediction + tools.

Through the lens | once you see it, you can't unsee it



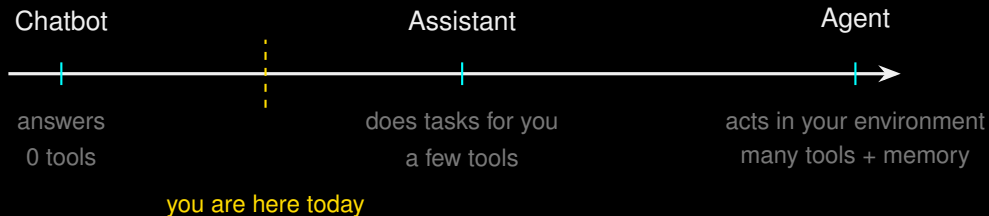
- ◇ The **lens**: look at any AI through it.
- ◇ **Agents**? next-token prediction + tools.
- ◇ **Alignment**? steering the same predictor.

Through the lens | once you see it, you can't unsee it

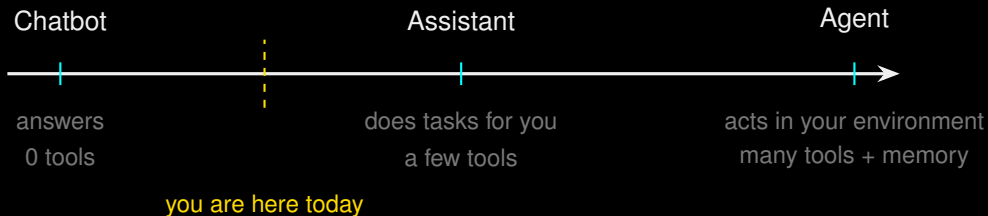


- ◇ The **lens**: look at any AI through it.
- ◇ **Agents**? next-token prediction + tools.
- ◇ **Alignment**? steering the same predictor.
- ◇ The hype is loud; the mechanism is simple.

The assistant axis | from chatbot to agent



The assistant axis | from chatbot to agent



Question: where do you want to live on this axis by the end of the year?

The alignment wall | a tiny corner of a huge research effort

Real research is happening on **AI alignment** — nine papers I keep going back to:

Concrete Problems in AI Safety

Amodei, Olah, et al.

2016 · [arXiv:1606.06565](#)

AI Safety via Debate

Irving, Christiano, Amodei

2018 · [arXiv:1805.00899](#)

Scalable Agent Alignment via Reward Modeling

Leike, Krueger, et al.

2018 · [arXiv:1811.07871](#)

Fine-Tuning LMs from Human Preferences

Ziegler, Stiennon, et al.

2019 · [arXiv:1909.08593](#)

Training a Helpful & Harmless Assistant (RLHF)

Bai, Jones, et al.

2022 · [arXiv:2204.05862](#)

Constitutional AI: Harmlessness from AI Feedback

Bai, Kadavath, et al.

2022 · [arXiv:2212.08073](#)

Towards Understanding Syco- phancy

Sharma, Tong, et al.

2023 · [arXiv:2310.13548](#)

Weak-to-Strong Generalization

Burns, Izmailov, et al.

2023 · [arXiv:2312.09390](#)

The AI Alignment Paradox

West & Aydin

2024 · [arXiv:2405.20806](#)

every paper is free on arXiv — pick one, read it tonight

The paper that started this fourth wave | Attention Is All You Need



Computer Science > Computation and Language

[Submitted on 12 Jun 2017 (v1), last revised 2 Aug 2023 (this version, v7)]

Attention Is All You Need

[Ashish Vaswani](#), [Noam Shazeer](#), [Niki Parmar](#), [Jakob Uszkoreit](#), [Llion Jones](#), [Aidan N. Gomez](#), [Lukasz Kaiser](#), [Illia Polosukhin](#)

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Comments: 15 pages, 5 figures

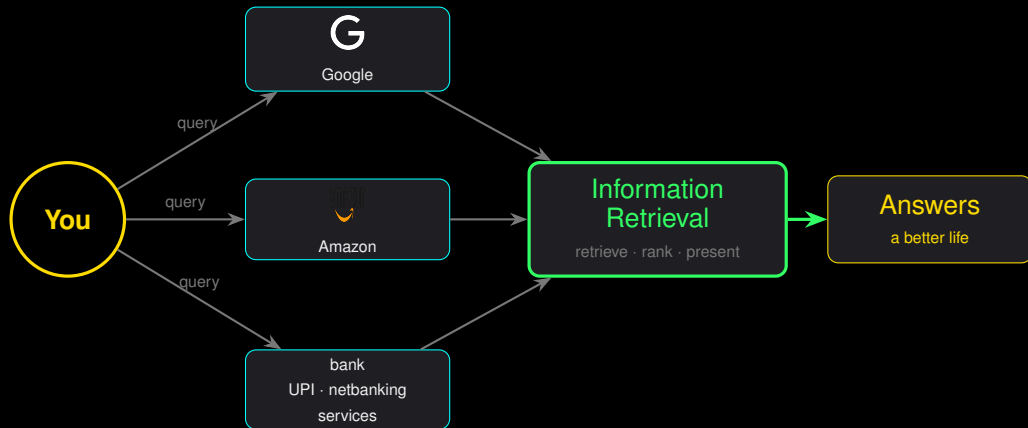
Subjects: **Computation and Language (cs.CL)**; Machine Learning (cs.LG)

Cite as: [arXiv:1706.03762](#) [cs.CL]
(or [arXiv:1706.03762v7](#) [cs.CL] for this version)
<https://doi.org/10.48550/arXiv.1706.03762> 

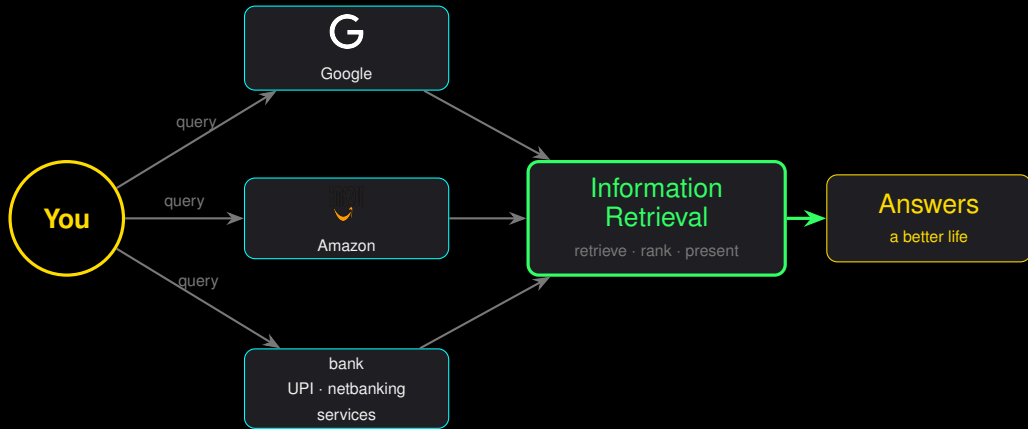
Vaswani et al., 2017 — **Attention Is All You Need**
arXiv:1706.03762

5. Fundamentals

How do you interact with computers? | spoiler: services



How do you interact with computers? | spoiler: services



Google, Amazon, even banks — they are **services**.
What do services do? **They retrieve information for you** — and make your life better.

Shuvam Banerji Seal

BS–MS final-year student, IISER Kolkata
Chemistry major · Computer Science minor
Research: **Information Retrieval** — RAG,
BM25, TREC, ECIR'26
Clubs: President, Slashdot · Office Bearer,
Valence

github.com/Shuvam-Banerji-Seal · shuvam-banerji-seal.github.io

- ◇ I study chemistry, I code for a living, I research **retrieval**.

Shuvam Banerji Seal

BS–MS final-year student, IISER Kolkata
Chemistry major · Computer Science minor
Research: **Information Retrieval** — RAG,
BM25, TREC, ECIR'26
Clubs: President, Slashdot · Office Bearer,
Valence

github.com/Shuvam-Banerji-Seal · shuvam-banerji-seal.github.io

- ◇ I study chemistry, I code for a living, I research **retrieval**.
- ◇ I teach tools for fun — like this talk.

Shuvam Banerji Seal

BS–MS final-year student, IISER Kolkata
Chemistry major · Computer Science minor
Research: **Information Retrieval** — RAG,
BM25, TREC, ECIR'26
Clubs: President, Slashdot · Office Bearer,
Valence

github.com/Shuvam-Banerji-Seal · shuvam-banerji-seal.github.io

- ◇ I study chemistry, I code for a living, I research **retrieval**.
- ◇ I teach tools for fun — like this talk.
- ◇ Everything I show you, I use daily.

You tell me.

You tell me.

?



Video

most of the bytes



Photos

many bytes

W

Text

few bytes, most value

You tell me.

?



Video

most of the bytes



Photos

many bytes



Text

few bytes, most value

out of the three — **text** is the one that constitutes all these services and softwares.

It's all text | the universal interface

- ◇ Out of **video**, **photos** and **text** — which one runs every service?



It's all text | the universal interface

- ◇ Out of **video**, **photos** and **text** — which one runs every service?
- ◇ **Text**. Text is the universal interface.



It's all text | the universal interface

- ◇ Out of **video**, **photos** and **text** — which one runs every service?
- ◇ **Text**. Text is the universal interface.
- ◇ Google? A text-index of the web. Your bank? Text rows in a database.



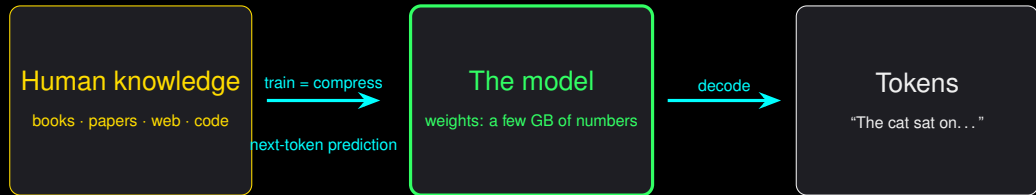
It's all text | the universal interface

- ◇ Out of **video**, **photos** and **text** — which one runs every service?
- ◇ **Text**. Text is the universal interface.
- ◇ Google? A text-index of the web. Your bank? Text rows in a database.
- ◇ LLMs eat text. **Agents act on text.**



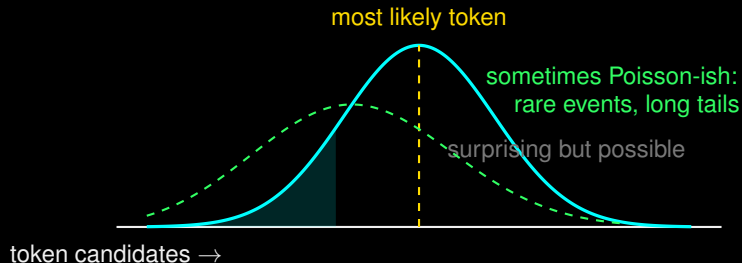
6. Knowledge Compression

LLMs = human knowledge, compressed into a model | the library in a file



A whole library, compressed into one file — that predicts the next word.

The distribution of the next token | a Gaussian (or maybe a Poisson)



The model is a **probability distribution** over the next token — your knowledge, your language, your style, compressed.

Too much maths? | okay, that was the last formula



- ◇ Don't worry — that was the only formula in this talk.

Too much maths? | okay, that was the last formula



- ◇ Don't worry — that was the only formula in this talk.
- ◇ The takeaway is one line: **LLMs predict the next token.**

Too much maths? | okay, that was the last formula



- ◇ Don't worry — that was the only formula in this talk.
- ◇ The takeaway is one line: **LLMs predict the next token.**
- ◇ And now that you know text matters. . .

Too much maths? | okay, that was the last formula



- ◇ Don't worry — that was the only formula in this talk.
- ◇ The takeaway is one line: **LLMs predict the next token.**
- ◇ And now that you know text matters. . .
- ◇ **let's get back to the tools.**

7. Tools You Should Learn

How many of you have a personal website? | hands up!

Hands up!

How many of you have a personal website? | hands up!

Hands up!

Want one? Host it free on:



GitHub



GitLab



Codeberg

Underlying tool: **git**. You need to learn git.

Hands up!

Want one? Host it free on:



GitHub



GitLab



Codeberg

Underlying tool: **git**. You need to learn git.

\$ your first repo

```
$ git init
$ git add .
$ git commit -m "my first website"
$ git push origin main
```


How many of you have a personal website? | hands up!

Hands up!

Want one? Host it free on:



GitHub



GitLab

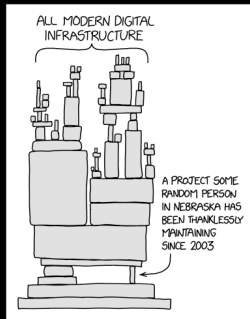


Codeberg

Underlying tool: **git**. You need to learn git.

\$ your first repo

```
$ git init
$ git add .
$ git commit -m "my first website"
$ git push origin main
```



xkcd 2347: Dependency — everyone runs on git, eventually

GitHub, in action | the project overview

Shuvam-Banerji-Seal / ai-in-real-world-context

Type to search

Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

ai-in-real-world-context

Public

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

<>

Code

Shuvam-Banerji-Seal

HTML: move xkcd Dependency meme from git slide to the O...

7b95f76 · 9 minutes ago

8 Commits

assets

Add 'Attention Is All You Need' slide (HTML+LaTeX); growl...

17 minutes ago

sections

Add 'Attention Is All You Need' slide (HTML+LaTeX); growl...

17 minutes ago

.gitignore

Fix step reveal (hide points until \pause-step), fresh mem...

3 hours ago

.olignore

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

LICENSE

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

README.md

Fix step reveal (hide points until \pause-step), fresh mem...

3 hours ago

index.html

HTML: move xkcd Dependency meme from git slide to th...

9 minutes ago

log

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

main.pdf

Add 'Attention Is All You Need' slide (HTML+LaTeX); growl...

17 minutes ago

main.tex

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

sbs_dark_beamer.sty

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

About

AI in Real World Context — a 60-min pointer-friendly HTML deck + LaTeX Beamer (tools, services, next-token predictors).

shuvam-banerji-seal.github.io/ai-l...

Readme

MIT license

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Deployments 8

github-pages 9 minutes ago

- ◇ PDFs: assignments, thesis, projects — all L^AT_EX.



Overleaf



L^AT_EX



Overleaf



L^AT_EX

- ◇ PDFs: assignments, thesis, projects — all L^AT_EX.
- ◇ You type text + commands; the machine typesets beauty.



Overleaf



L^AT_EX

- ◇ PDFs: assignments, thesis, projects — all L^AT_EX.
- ◇ You type text + commands; the machine typesets beauty.
- ◇ Overleaf = L^AT_EX in your browser, zero install.



Overleaf



L^AT_EX

- ◇ PDFs: assignments, thesis, projects — all L^AT_EX.
- ◇ You type text + commands; the machine typesets beauty.
- ◇ Overleaf = L^AT_EX in your browser, zero install.

fun fact

Heck — even this presentation is made using L^AT_EX.
You are literally reading typeset tokens.

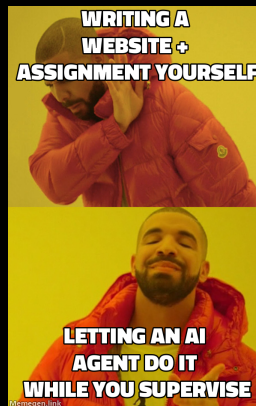
Overleaf, in action | the project view

The screenshot displays the Overleaf web editor interface. The top navigation bar includes 'File', 'Edit', 'Insert', 'View', 'Format', 'Help', and an 'Upgrade' button. The main title bar reads 'Machine Verified Proving of Biological Information'. The left sidebar contains a 'File tree' and a 'File outline'. The 'File tree' shows a project structure with sections like '01_prologue.tex', '02_agenda.tex', '03_tooltime.tex', '04_aillandscape.tex', '05_fundamentals.tex', '06_knowledge.tex' (highlighted), '07_tools.tex', and '08_agents.tex'. The 'File outline' shows a hierarchy for 'Knowledge Compression' with sub-items like 'Human Knowledge -> ...', 'LLMs - human kn...', 'The Distribution', 'The distribution of...', and 'Too Much Maths?'. The main editor area shows LaTeX code for a document section titled 'Too much maths?'. The code includes a frame, a subsection, and a list of items. The right sidebar shows a preview of the document, which includes the Overleaf logo and a list of PDFs: 'assignments, thesis, projects — all' and 'MTeX'.

Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

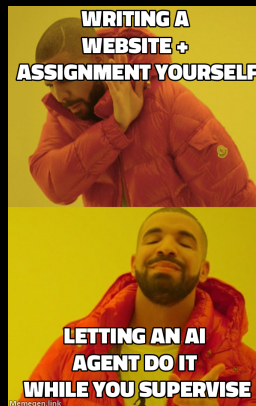
- ◇ Would you want to do it all **by yourself**?



Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

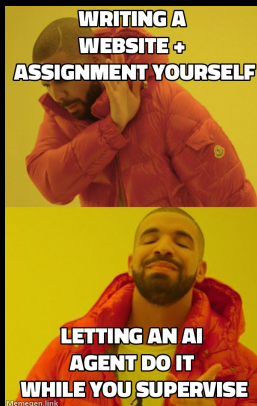
- ◇ Would you want to do it all **by yourself**?
- ◇ Remember: everything here is **text**.



Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

- ◇ Would you want to do it all **by yourself**?
- ◇ Remember: everything here is **text**.
- ◇ And text is exactly what AI agents are good at.



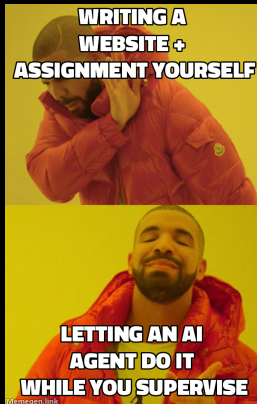
Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

- ◇ Would you want to do it all **by yourself**?
- ◇ Remember: everything here is **text**.
- ◇ And text is exactly what AI agents are good at.

the pitch

Use **agentic tools** — like the ones I'm about to show.



8. Agentic Tools

Use agentic tools | AI that acts, not just answers

- ◇ Tools I use daily: **VS Code** + **agentic coding tools**.

a session, roughly

```
$ opencode
» write a python script that
» downloads all my course PDFs
✓ plan:  fetch, parse, save
✓ wrote downloads.py
✓ ran it:  42 files saved
$
```

Use agentic tools | AI that acts, not just answers

- ◇ Tools I use daily: **VS Code** + **agentic coding tools**.
- ◇ Open-source, runnable locally: OpenCode, Claude Code, ...

a session, roughly

```
$ opencode
» write a python script that
» downloads all my course PDFs
✓ plan:  fetch, parse, save
✓ wrote downloads.py
✓ ran it:  42 files saved
$
```

Use agentic tools | AI that acts, not just answers

- ◇ Tools I use daily: **VS Code** + **agentic coding tools**.
- ◇ Open-source, runnable locally: OpenCode, Claude Code, ...
- ◇ They read your files, write code, run commands — you **supervise**.

a session, roughly

```
$ opencode
» write a python script that
» downloads all my course PDFs
✓ plan:  fetch, parse, save
✓ wrote downloads.py
✓ ran it:  42 files saved
$
```

not sci-fi — this is a Tuesday.

Use agentic tools | AI that acts, not just answers

- ◇ Tools I use daily: **VS Code** + **agentic coding tools**.
- ◇ Open-source, runnable locally: OpenCode, Claude Code, ...
- ◇ They read your files, write code, run commands — you **supervise**.
- ◇ This presentation? Written by me **with** an agent.

a session, roughly

```
$ opencode
» write a python script that
» downloads all my course PDFs
✓ plan:  fetch, parse, save
✓ wrote downloads.py
✓ ran it:  42 files saved
$
```

not sci-fi — this is a Tuesday.

Use agentic tools | AI that acts, not just answers

- Tools I use daily: **VS Code** + **agentic coding tools**.
- Open-source, runnable locally: OpenCode, Claude Code, ...
- They read your files, write code, run commands — you **supervise**.
- This presentation? Written by me **with** an agent.



OpenAI



ChatGPT



Claude



Gemini



HF



shell

a session, roughly

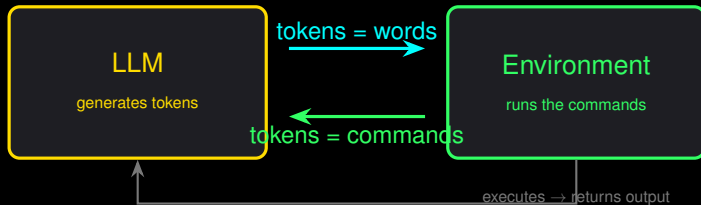
```
$ opencode
» write a python script that
» downloads all my course PDFs
✓ plan:  fetch, parse, save
✓ wrote downloads.py
✓ ran it:  42 files saved
$
```

not sci-fi — this is a Tuesday.

What are agentic tools? | tokens become commands



What are agentic tools? | tokens become commands



LLMs generate tokens (words). What if the token is a **command** — and the environment it runs in **executes the command** as well? That's an **agentic tool**.

Questions? Everything is text.

Agents turn text into action.

github.com/Shuvam-Banerji-Seal · shuvam-banerji-seal.github.io
